In this problem, you will manually compute the forward inference of a small neural network. The goal of this exercise is to develop a clear understanding of how fully connected neural networks perform numerical computations, including linear transformations, nonlinear activations, softmax normalization, and cross-entropy loss.

Network Architecture

The network consists of the following components:

- A 5-dimensional input vector.
- A hidden layer with ReLU activation.
- An output layer that produces logits for a 3-class classification problem.

Input

The input to the network is a vector

$$x \in \mathbb{R}^5.$$

First Layer (Hidden Layer)

The first layer performs a linear transformation followed by a ReLU activation:

$$a = W_1 x + b_1, \quad h = \mathrm{ReLU}(a),$$

where the ReLU function is applied element-wise:

$$\mathrm{ReLU}(t) = \max(0, t).$$

The parameters of the first layer are:

$$W_1 \in \mathbb{R}^{3\times 5}, \quad b_1 \in \mathbb{R}^3.$$

Second Layer (Output Layer)

The second layer produces the logits:

$$z = W_2 h + b_2,$$

where

$$W_2 \in \mathbb{R}^{3\times 3}, \quad b_2 \in \mathbb{R}^3.$$

Softmax and Loss

The logits are converted to class probabilities using the softmax function:

$$p_i = \frac{e^{z_i}}{\sum_{k=1}^{3} e^{z_k}}, \quad i = 1, 2, 3.$$

The true label for the given input sample is **class 3**. The cross-entropy loss for a single sample is defined as:

$$L = -\ln p_3.$$

Given Data

Input Vector

$$x = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

First Layer Parameters

$$W_1 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

Second Layer Parameters

$$W_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Questions

Q1

Compute the first-layer linear output

$$a = W_1 x + b_1.$$

Q2

Compute the hidden-layer output after applying the ReLU activation:

$$h = \mathrm{ReLU}(a).$$

Q3

Compute the logits:

$$z = W_2 h + b_2.$$

Q4

Compute the softmax probabilities. Report all probabilities rounded to **three decimal places**.

Q5

Compute the cross-entropy loss. Report the loss rounded to **three decimal places**.

$$L = -\ln p_3.$$